IMPLEMENTATION OF INTEGRITY CONSTRAINTS

ASSIGNMENT 6

DATABASE MANAGEMENT SYSTEM

ALLIANCE UNIVERSITY

Department of Computer Applications

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COURSE: BCA

YEAR: 2ND YEAR

Introduction

Integrity constraints are rules applied to database tables to maintain the accuracy, validity, and consistency of data. SQL constraints help prevent invalid or duplicate data from being entered into a database. In this assignment, primary key, foreign key, UNIQUE, CHECK, and DEFAULT constraints are implemented in the Food Delivery Management System using MySQL.

1. Primary Key and Foreign Key Constraints

Primary keys uniquely identify each record in a table, whereas foreign keys establish relationships between tables and maintain referential integrity.

Table	Primary Key	Foreign Key
CUSTOMER	Customer_ID	-
RESTAURANT	Restaurant_ID	-
MENU_ITEM	Item_ID	Restaurant_ID
DELIVERY_PARTNER	Partner_ID	-
ORDERS	Order_ID	Customer_ID, Restaurant_ID, Partner_ID
ORDER_ITEM	Order_Item_ID	Order_ID, Item_ID
PAYMENT	Payment_ID	Order_ID

Primary keys prevent duplicate identifiers and NULL primary key values. Foreign keys ensure that related records refer to existing records in the parent tables.

2. Implementing UNIQUE, CHECK, and DEFAULT Constraints

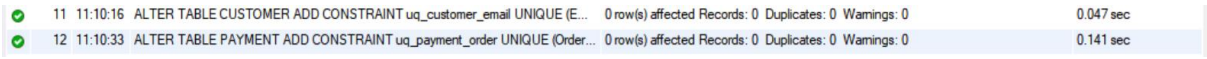
A. UNIQUE Constraint

The UNIQUE constraint prevents duplicate values in a column. It can be used to ensure that customer email addresses are not repeated and that each order has only one payment record.

```
USE Food_Delivery_DB;


ALTER TABLE CUSTOMER
ADD CONSTRAINT uq_customer_email UNIQUE (Email);
```

```
ALTER TABLE PAYMENT
ADD CONSTRAINT uq_payment_order UNIQUE (Order_ID);
```



- uq_customer_email prevents two customers from having the same email address.
- uq_payment_order ensures that an order can have only one payment record, enforcing the intended one-to-one relationship between ORDERS and PAYMENT.

B. CHECK Constraint

The CHECK constraint restricts the values that can be stored in a column.

```
ALTER TABLE RESTAURANT
ADD CONSTRAINT chk_restaurant_rating
CHECK (Rating >= 0 AND Rating <= 5);

ALTER TABLE MENU_ITEM
ADD CONSTRAINT chk_menu_price
CHECK (Price >= 0);

ALTER TABLE ORDER_ITEM
ADD CONSTRAINT chk_item_quantity
CHECK (Quantity > 0);

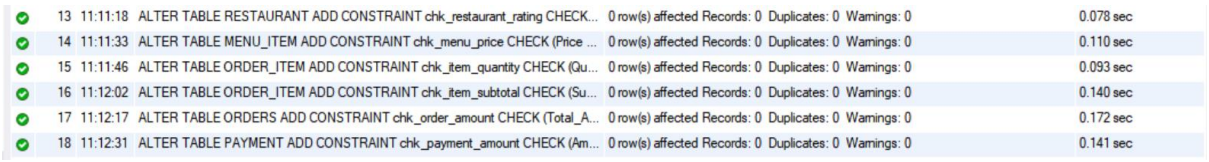
ALTER TABLE ORDER_ITEM
ADD CONSTRAINT chk_item_subtotal
CHECK (Subtotal >= 0);

ALTER TABLE ORDERS
ADD CONSTRAINT chk_order_amount
CHECK (Total_Amount >= 0);
```

ALTER TABLE PAYMENT

ADD CONSTRAINT chk_payment_amount

CHECK (Amount >= 0);



13	11:11:18	ALTER TABLE RESTAURANT ADD CONSTRAINT chk_restaurant_rating CHECK...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.078 sec
14	11:11:33	ALTER TABLE MENU_ITEM ADD CONSTRAINT chk_menu_price CHECK (Price ...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.110 sec
15	11:11:46	ALTER TABLE ORDER_ITEM ADD CONSTRAINT chk_item_quantity CHECK (Qu...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.093 sec
16	11:12:02	ALTER TABLE ORDER_ITEM ADD CONSTRAINT chk_item_subtotal CHECK (Su...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.140 sec
17	11:12:17	ALTER TABLE ORDERS ADD CONSTRAINT chk_order_amount CHECK (Total_A...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.172 sec
18	11:12:31	ALTER TABLE PAYMENT ADD CONSTRAINT chk_payment_amount CHECK (Am...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.141 sec

- Restaurant ratings must be between 0 and 5.
- Menu item prices and order amounts cannot be negative.
- Order item quantities must be greater than zero.
- Payment amounts cannot be negative.

C. DEFAULT Constraint

The DEFAULT constraint assigns a value automatically when a record is inserted without specifying a value for that column.

ALTER TABLE MENU_ITEM

ALTER COLUMN Availability SET DEFAULT TRUE;

ALTER TABLE ORDERS

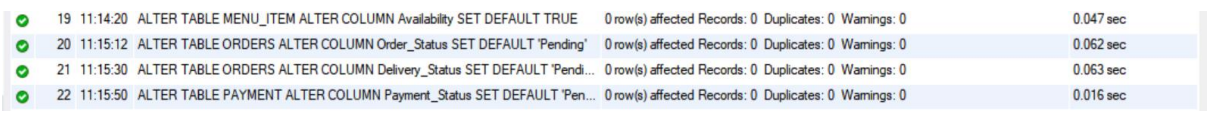
ALTER COLUMN Order_Status SET DEFAULT 'Pending';

ALTER TABLE ORDERS

ALTER COLUMN Delivery_Status SET DEFAULT 'Pending';

ALTER TABLE PAYMENT

ALTER COLUMN Payment_Status SET DEFAULT 'Pending';



19	11:14:20	ALTER TABLE MENU_ITEM ALTER COLUMN Availability SET DEFAULT TRUE	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.047 sec
20	11:15:12	ALTER TABLE ORDERS ALTER COLUMN Order_Status SET DEFAULT 'Pending'	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.062 sec
21	11:15:30	ALTER TABLE ORDERS ALTER COLUMN Delivery_Status SET DEFAULT 'Pendi...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.063 sec
22	11:15:50	ALTER TABLE PAYMENT ALTER COLUMN Payment_Status SET DEFAULT 'Pen...	0 row(s) affected Records: 0	Duplicates: 0	Warnings: 0	0.016 sec

If a value is omitted during insertion, MySQL uses the specified default value for the column. These statements use MySQL 8.0.13 or later syntax.

3. Testing the Constraints

The following commands can be used to verify that the constraints have been added.

- SHOW CREATE TABLE CUSTOMER;
- SHOW CREATE TABLE RESTAURANT;
- SHOW CREATE TABLE MENU_ITEM;
- SHOW CREATE TABLE ORDERS;
- SHOW CREATE TABLE ORDER_ITEM;
- SHOW CREATE TABLE PAYMENT;

The output displays the table definitions, including the constraints.

To check the default value of the menu item availability column, execute:

- SHOW COLUMNS FROM MENU_ITEM;

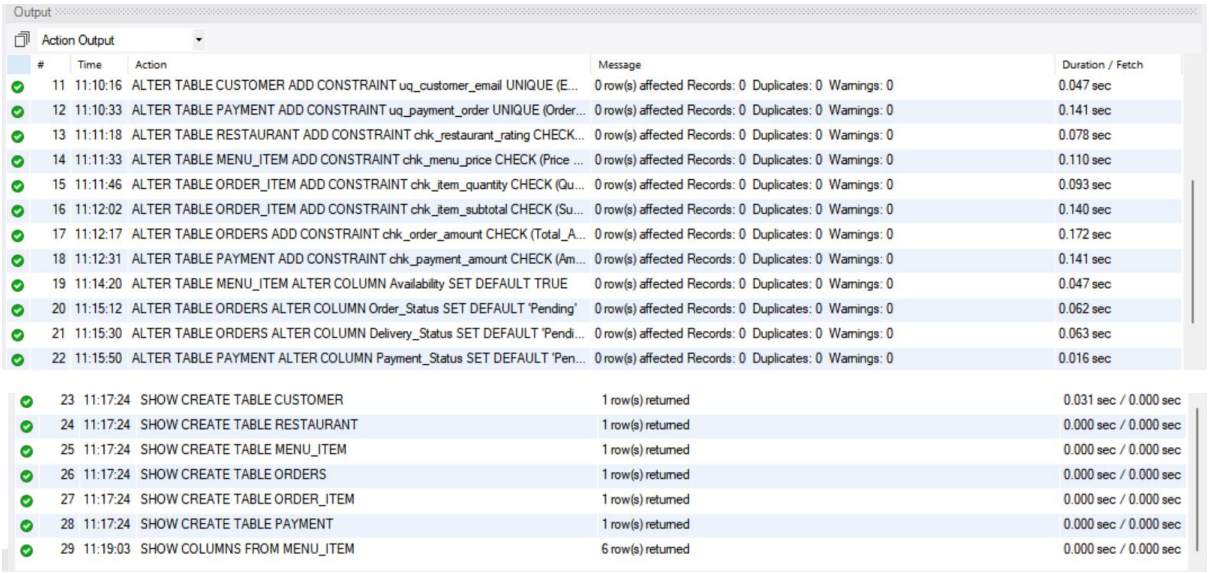
Field	Type	Null	Key	Default	Extra
Item_ID	int	NO	PRI	NULL	
Restaurant_ID	int	YES	MUL	NULL	
Item_Name	varchar(100)	YES		NULL	
Category	varchar(50)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Availability	tinyint(1)	YES		1	

The Default column should display 1 or TRUE, depending on the MySQL interface.

Expected result: The database tables contain the specified UNIQUE, CHECK, and DEFAULT constraints. Invalid values, duplicate unique values, and duplicate payment records for the same order are restricted by the corresponding constraints.

Conclusion

Integrity constraints improve the reliability of the Food Delivery Management System database. Primary keys uniquely identify records, foreign keys maintain relationships, UNIQUE constraints prevent duplicate values, CHECK constraints restrict invalid data, and DEFAULT constraints automatically provide predefined values. Together, these constraints help maintain data accuracy and consistency.



Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	11:10:16	ALTER TABLE CUSTOMER ADD CONSTRAINT uq_customer_email UNIQUE (E...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
12	11:10:33	ALTER TABLE PAYMENT ADD CONSTRAINT uq_payment_order UNIQUE (Order...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.141 sec
13	11:11:18	ALTER TABLE RESTAURANT ADD CONSTRAINT chk_restaurant_rating CHECK...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
14	11:11:33	ALTER TABLE MENU_ITEM ADD CONSTRAINT chk_menu_price CHECK (Price ...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.110 sec
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22	11:15:50	ALTER TABLE PAYMENT ALTER COLUMN Payment_Status SET DEFAULT 'Pen...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
23	11:17:24	SHOW CREATE TABLE CUSTOMER	1 row(s) returned	0.031 sec / 0.000 sec
24	11:17:24	SHOW CREATE TABLE RESTAURANT	1 row(s) returned	0.000 sec / 0.000 sec
25	11:17:24	SHOW CREATE TABLE MENU_ITEM	1 row(s) returned	0.000 sec / 0.000 sec
26	11:17:24	SHOW CREATE TABLE ORDERS	1 row(s) returned	0.000 sec / 0.000 sec
27	11:17:24	SHOW CREATE TABLE ORDER_ITEM	1 row(s) returned	0.000 sec / 0.000 sec
28	11:17:24	SHOW CREATE TABLE PAYMENT	1 row(s) returned	0.000 sec / 0.000 sec
29	11:19:03	SHOW COLUMNS FROM MENU_ITEM	6 row(s) returned	0.000 sec / 0.000 sec